

	Work Instruction	Doc. Revision	02
	AXI V810 STD X Axis Ball Screw Alignment Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	15 Jan 2013	First Release	ONG KEAN SHENG
01	04 Jan 2015	Second Release	Muhammed Shameem
02	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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1. Loosen the bearing the Bearing support with M6 Allen Key at “A” & “B” (Note: DO NOT TIGHTEN) as shown in Figure 1.
2. Loosen the Bearing support, Fixed round at M5 Allen Key at “G” to “J”. (Note: DO NOT TIGHTEN) as shown in Figure 1.
3. Loosen the screw at round bearing with M6 Allen Key at “C” to “F”. (Note: DO NOT TIGHTEN) as shown in Figure 1.

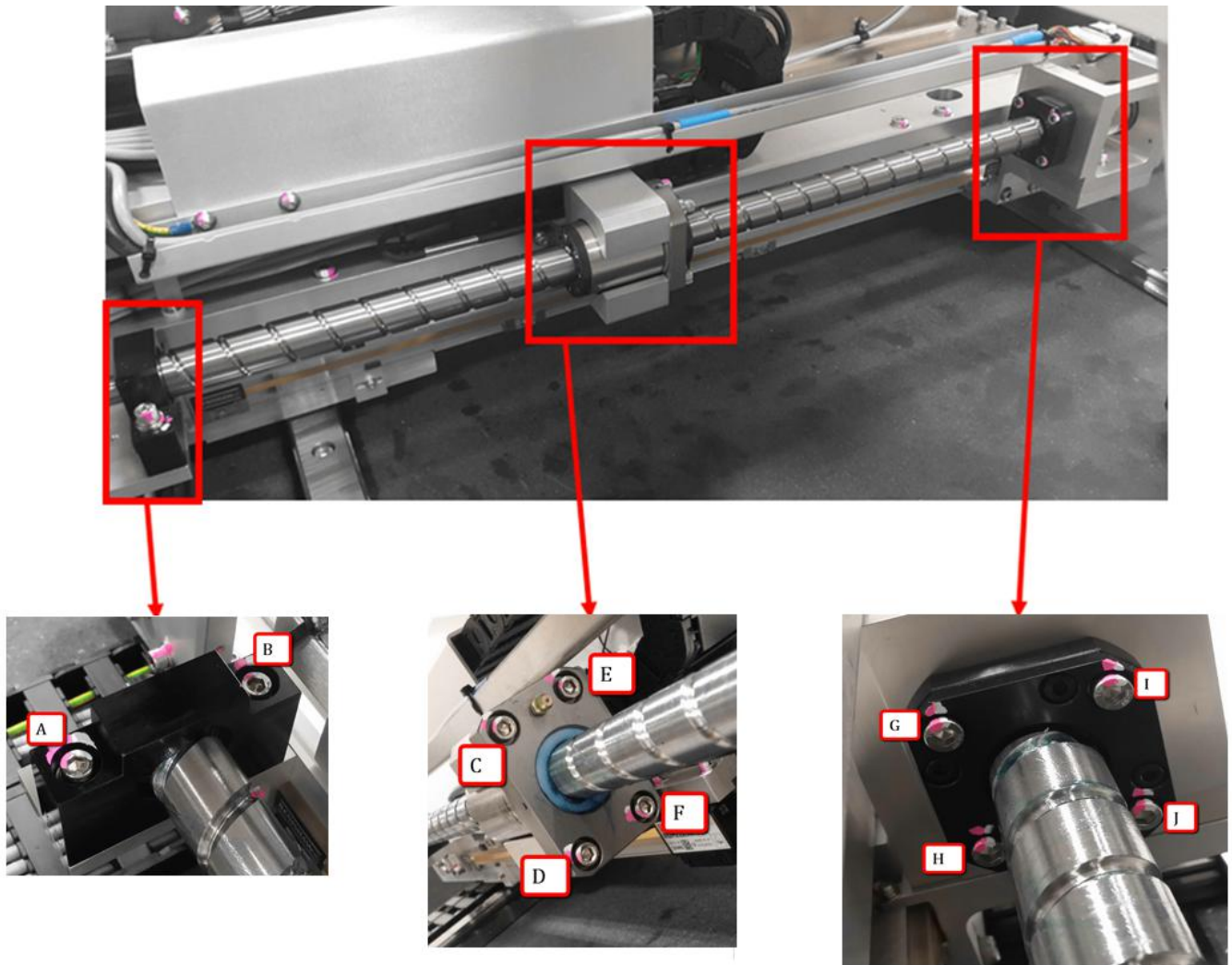


Figure 1 X-axis Ball Screw

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4. Loosen 2 screws at coupling (which join to the servo motor) as shown in Figure 2.

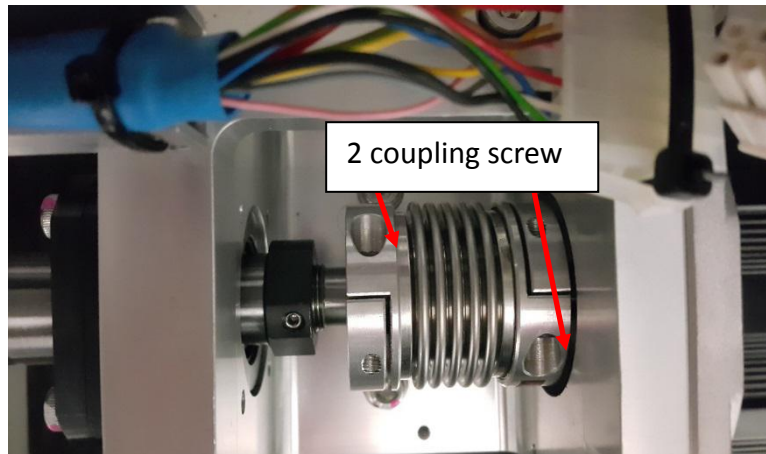


Figure 2 Coupling Screws

5. Loosen 2 screws in the Motor Mount as shown in the Figure 3.

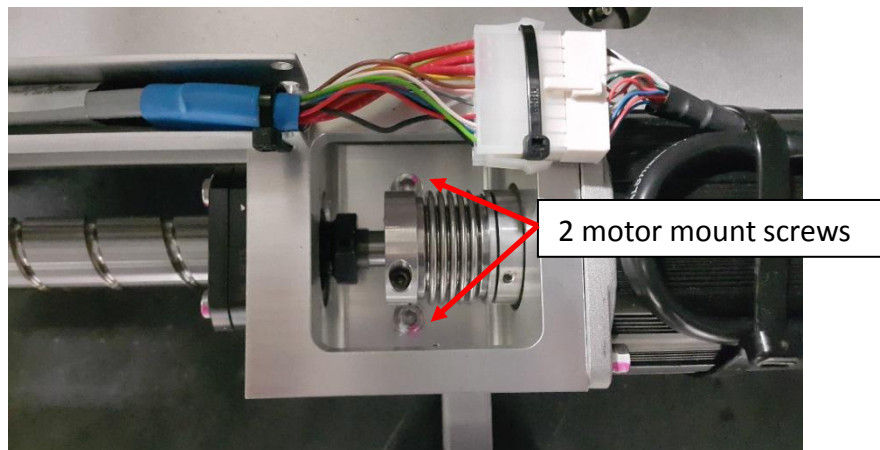


Figure 3 Motor Mount

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6. Push the stage from minimum and maximum hard stop(Left to Right)
7. Pull the stage from maximum to minimum hard stop(Right to Left)
8. Repeat Step 6 to 7 for 3 times.

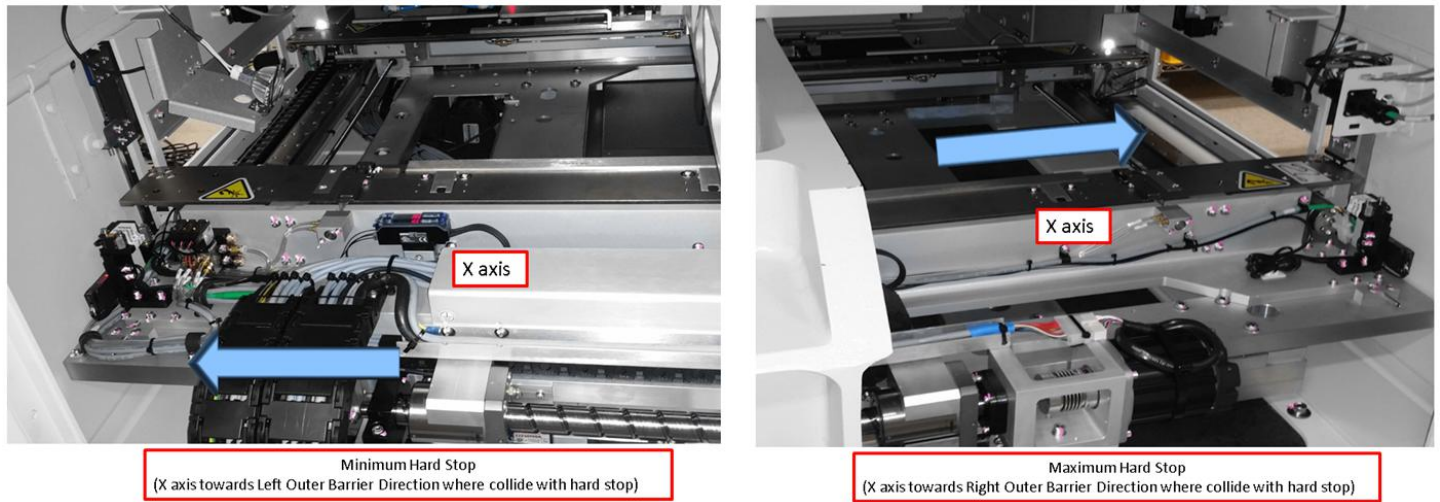


Figure 4 Minimum & Maximum Hard Stop (V810 Std)

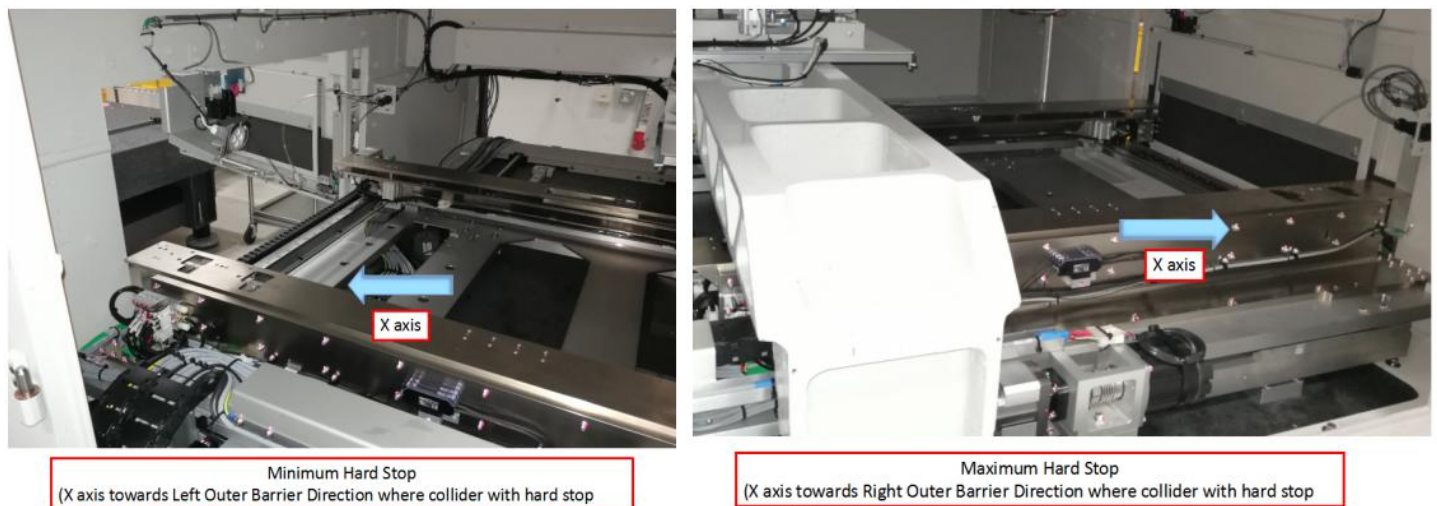


Figure 5 Minimum & Maximum Hard Stop (V810 XXL)

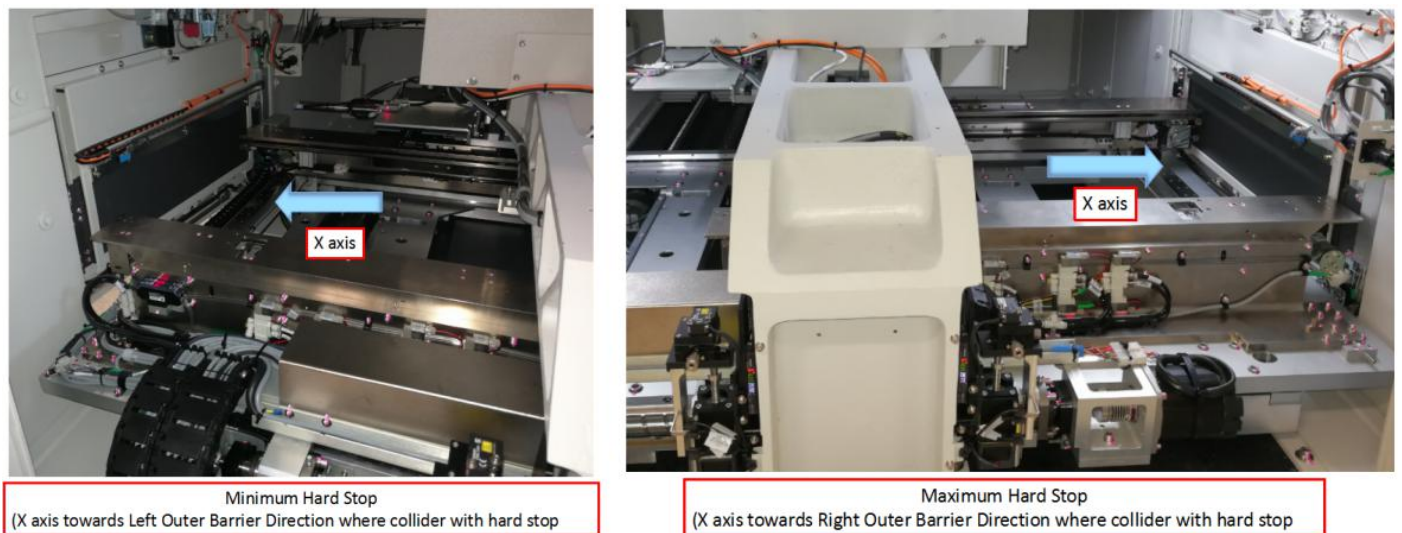


Figure 6 Minimum & Maximum Hard Stop (V810 S2EX)

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9. Push the X axis from Home position to the middle of the stage and tighten 4 screws at Bearing support, Fixed round as shown in Figure 7.

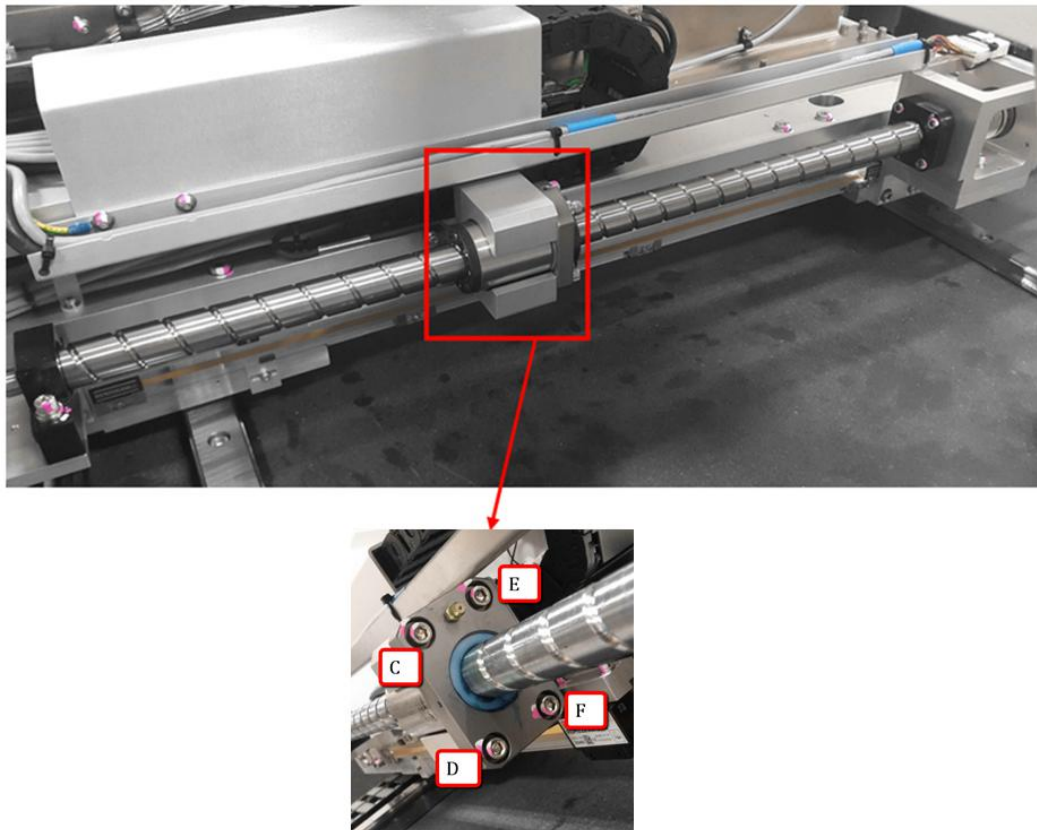
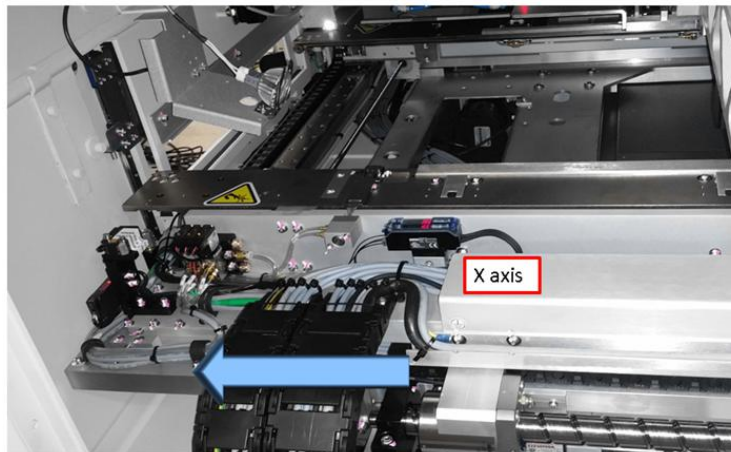


Figure 7 Middle Support Bearing

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10. Push the stage from minimum and maximum hard stop(Left to Right)
11. Pull the stage from maximum to minimum hard stop(Right to Left)
12. Repeat Step 10 to 11 for 3 times.



Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collide with hard stop)



Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collide with hard stop)

Figure 8 Minimum & Maximum Hard Stop (V810 Std)

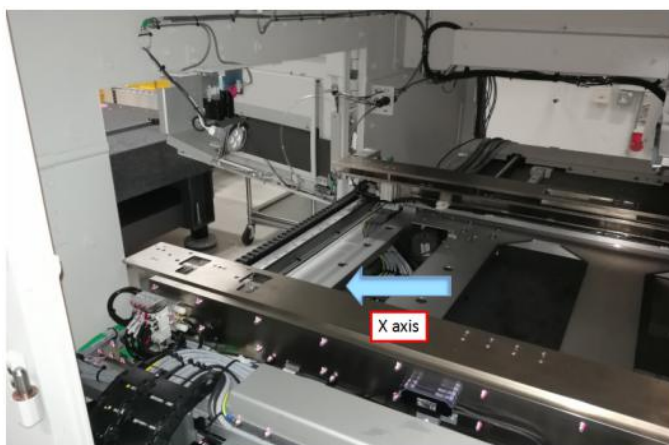


Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collider with hard stop)

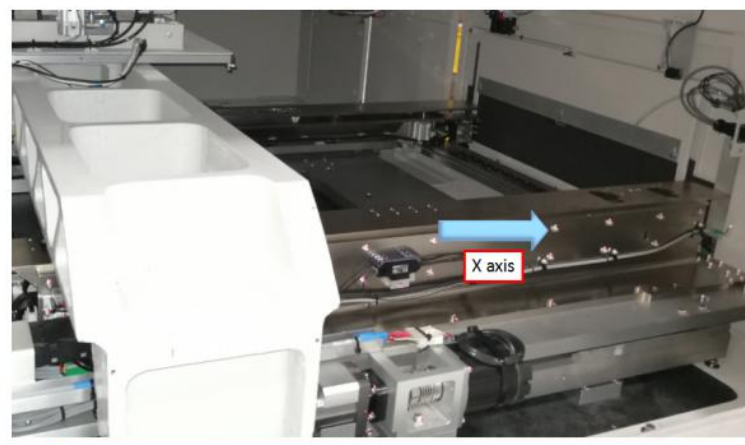


Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collider with hard stop)

Figure 9 Minimum & Maximum Hard Stop (V810 XXL)



Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collider with hard stop)



Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collider with hard stop)

Figure 10 Minimum & Maximum Hard Stop (V810 S2EX)

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13. Push the X axis stage to the maximum hard stop (leave a small space for your Allen key to access the Bearing Support, Fixed) & tighten 4 screws at Bearing support, Fixed round as shown in Figure 11.

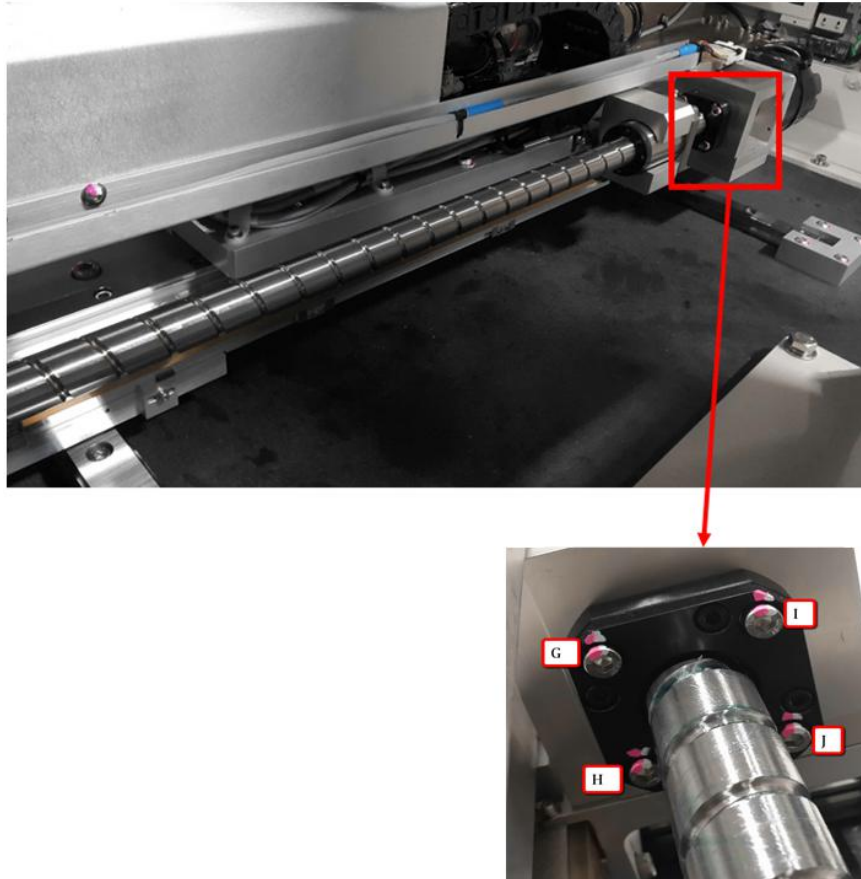
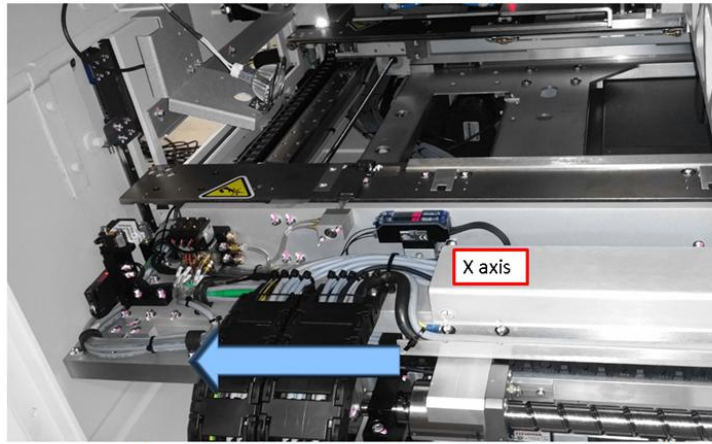


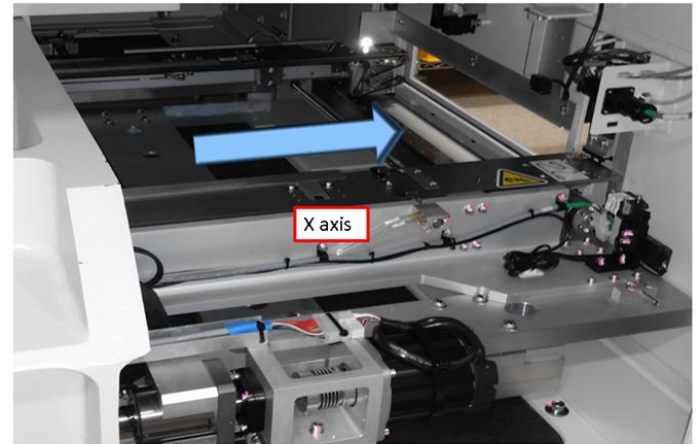
Figure 11 End support bearing

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14. Push the stage from minimum and maximum hard stop(Left to Right)
15. Pull the stage from maximum to minimum hard stop(Right to Left)
16. Repeat Step 14 to 15 for 3 times.



Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collide with hard stop)

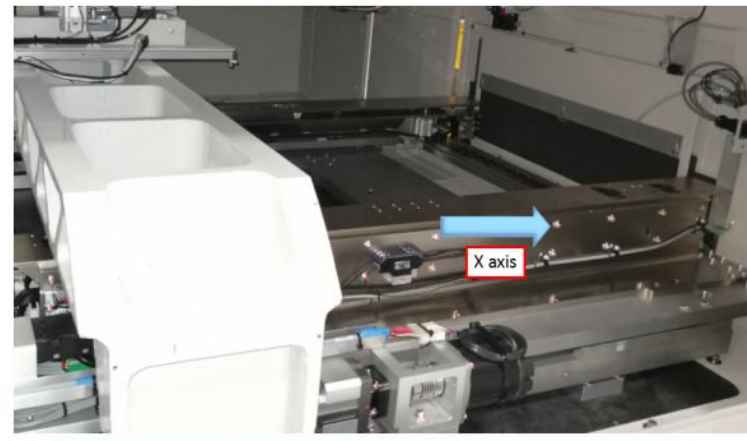


Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collide with hard stop)

Figure 12 Minimum & Maximum hard stop (V810 Std)



Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collider with hard stop)

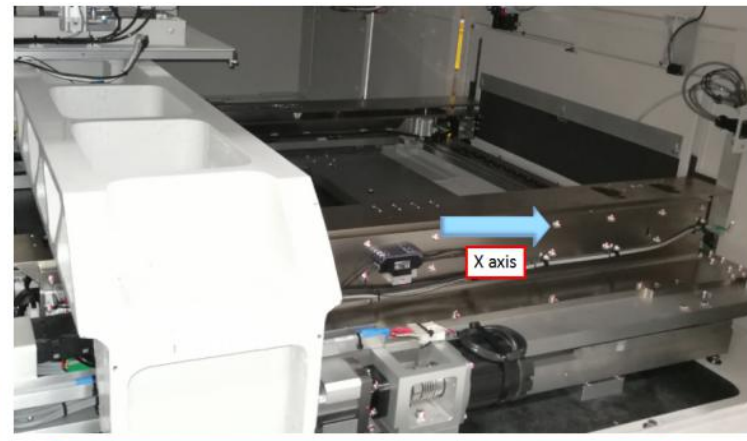


Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collider with hard stop)

Figure 13 Minimum & Maximum hard stop (V810 XXL)



Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collider with hard stop)



Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collider with hard stop)

Figure 14 Minimum & Maximum hard stop (V810 S2EX)

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17. Push the X-Axis Stage to minimum hard stop as shown in the Figure 15 and tighten screws at End support bearing

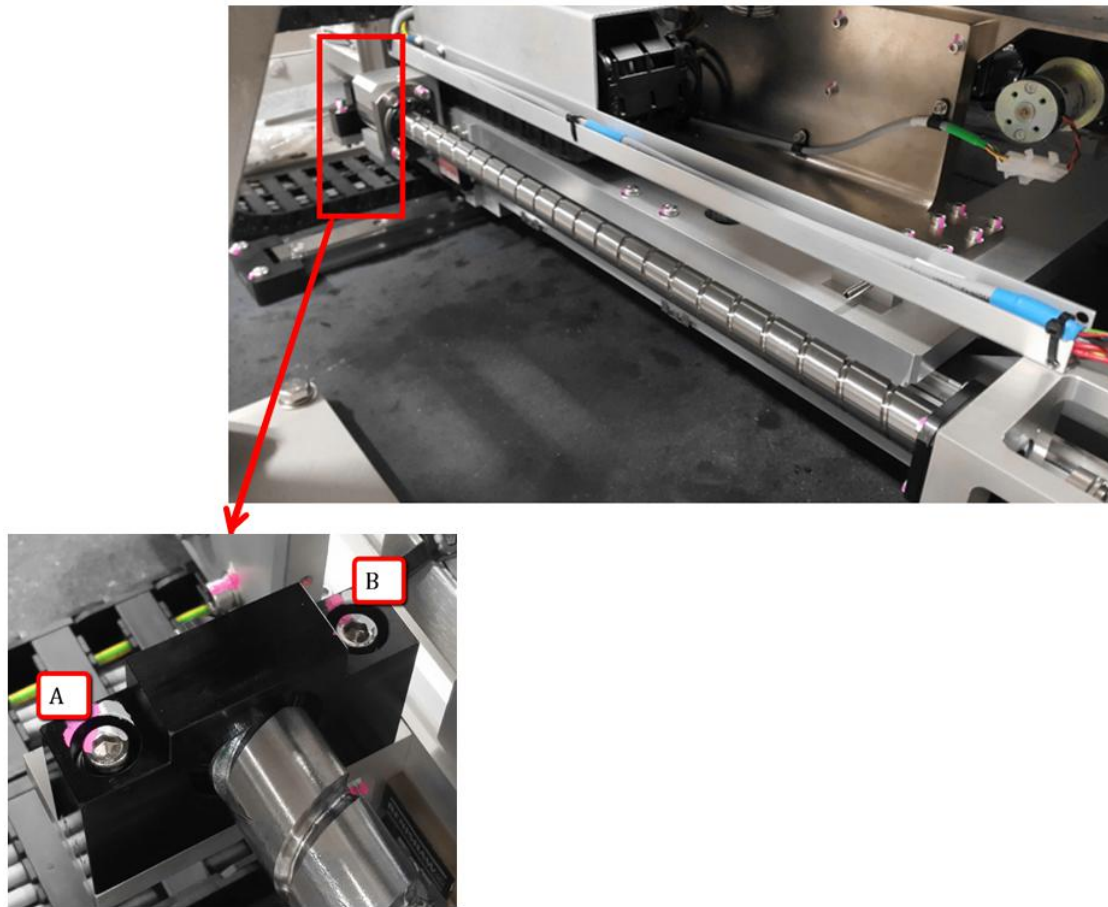
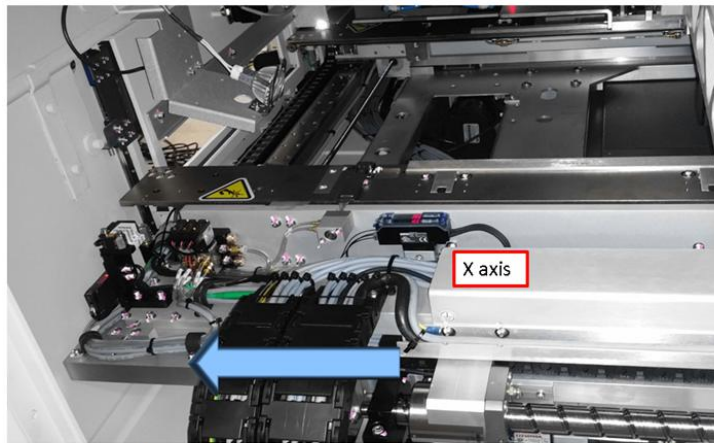


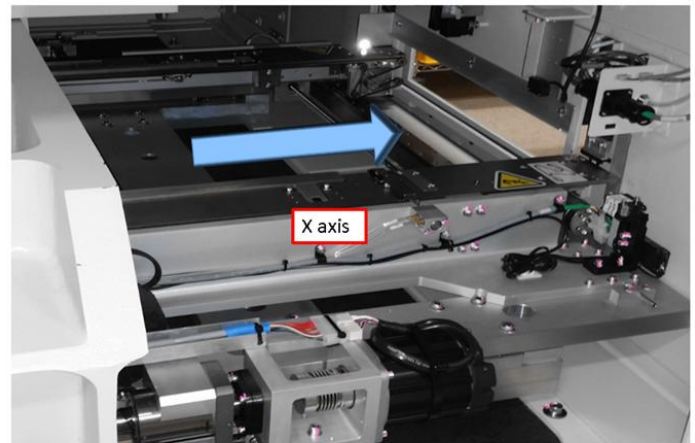
Figure 15 End support bearing (Left)

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18. Push the stage from minimum and maximum hard stop(Left to Right)
19. Pull the stage from maximum to minimum hard stop(Right to Left)
20. Repeat Step 18 to 19 for 3 times.

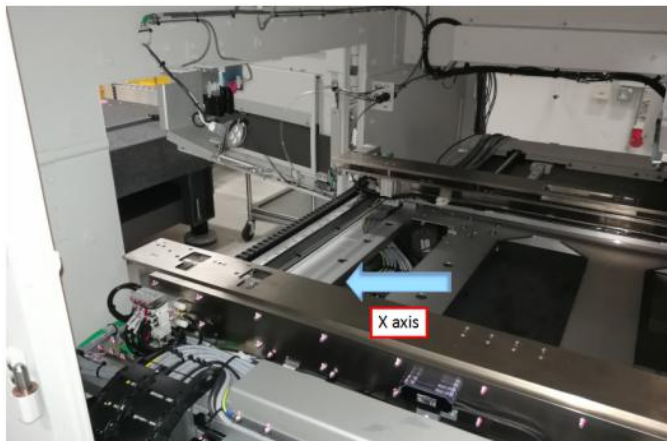


Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collide with hard stop)

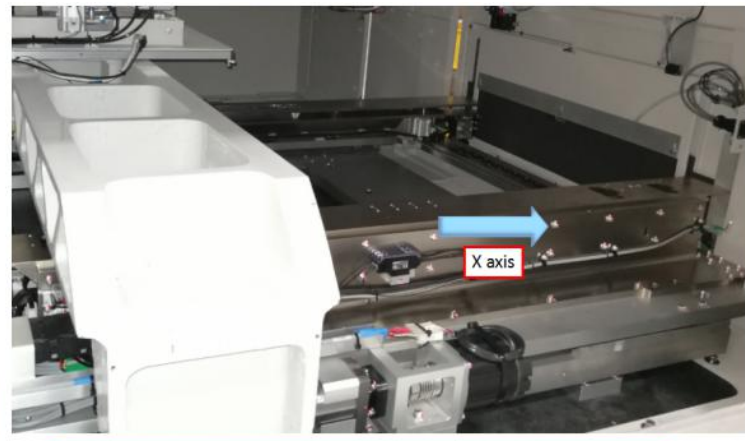


Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collide with hard stop)

Figure 16 Minimum & Maximum Hard Stop (V810 Std)

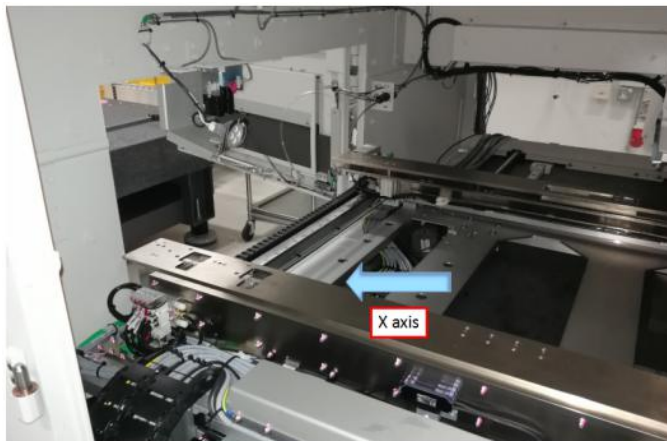


Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collider with hard stop)

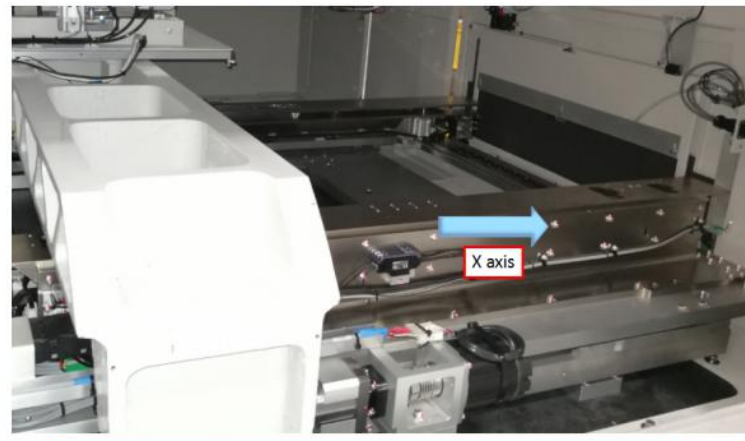


Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collider with hard stop)

Figure 17 Minimum & Maximum Hard Stop (V810 XXL)



Minimum Hard Stop
(X axis towards Left Outer Barrier Direction where collider with hard stop)



Maximum Hard Stop
(X axis towards Right Outer Barrier Direction where collider with hard stop)

Figure 18 Minimum & Maximum Hard Stop (V810 S2EX)

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21. Tighten back the 2 screws of the motor mount as below Figure 19.

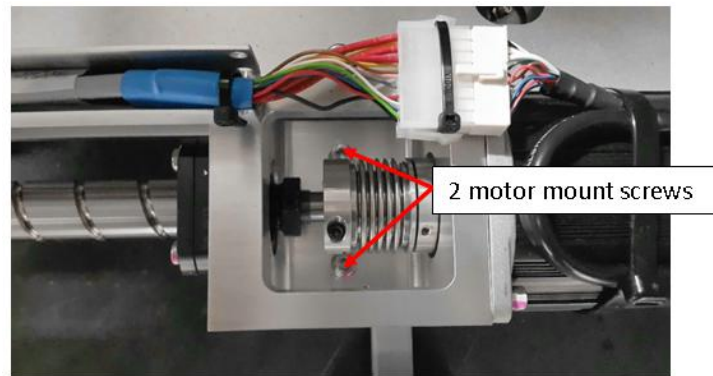


Figure 19 Motor Mount Screws

22. Tighten back 2 x Coupling Screws as below Figure 20.

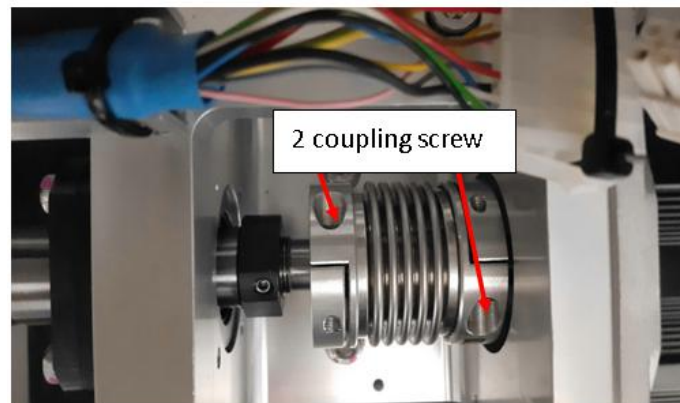


Figure 20 Coupling Screws

23. Push the stage to the maximum hard stop and close the front door.

24. Go to V810 software and log in accordingly.

25. Go to the Services->Subsystem Status & Control -> Panel Positioner and Click Initialize.

26. Run 5 Loops of Long Scan at Panel Positioner.

27. Run 1 loop of CD &A.

28. If X axis stage is encounter abnormality during step 26 Or 27, repeat Step 1 top Step 22.